

Aarogya Sahayak x Lyzr AI

Enterprise 4-Agent Consensus Mesh for Rural Indian Healthcare

Platform: Aarogya Sahayak | **Deployment:** Lyzr AI Studio (studio.lyzr.ai) | **Consensus Mesh:** 4 Autonomous Agents (Manager, Guardrail, Schemes, Protocols) | **Contact:** admin@aarogyasahayak.in

1. Executive Summary & Multi-Agent Architecture

In rural India, frontline Community Health Workers (ASHAs) provide primary triage for hundreds of millions of citizens. When an expectant mother presents with symptoms of **pre-eclampsia** (BP 160/100 mmHg with blurred vision and severe headache), relying on a single monolithic LLM prompt is dangerous: it frequently hallucinates dosages, misses MoHFW protocol danger signs, and provides zero safety auditability. By integrating **Lyzr AI**, Aarogya Sahayak implements an enterprise-grade **4-Agent Consensus Mesh** where specialized autonomous agents collaborate under rigorous clinical and safety guardrails.

2. Deployed Lyzr Multi-Agent Specification (All 4 Live)

Agent Role / Name	Lyzr Live Agent ID	Foundation Model	Core Mandate & Clinical Responsibility
1. Manager Agent (Aarogya Clinical Navigator)	6a9ae0e14a372650b843a9ae	OpenAI gpt-4o	Orchestrates patient intake, triages symptoms & vitals, delegates to safety guardrail.
2. Medical Safety Guardrail (Aarogya Safety Guardrail)	6a9ae9404e6f909d5b1ce8e7	OpenAI gpt-4o	Six-Sigma Clinical Auditor with absolute veto power. Blocks unauthorized drug prescribing.
3. Welfare Schemes Specialist (Aarogya Welfare Schemes)	6a9aeb88f70815409cbca57f	OpenAI gpt-4o	Calculates entitlements: Janani Suraksha Yojana (INR 1,400), PMMVY (INR 5,000 DBT), PM-JAY.
4. Clinical Protocols RAG (Aarogya Clinical Protocol)	6a9aec908d69d22325c3e67f	OpenAI gpt-5.4-mini	Grounds assessments in MoHFW maternal protocols & ICMR Standard Treatment Workflows.

3. Dual-Sponsor Synergy: Lyzr AI (The Brain) + Swytchcode (The Hand)

Aarogya Sahayak combines two industry sponsors into an end-to-end reliable agentic system:

- Lyzr AI (The Cognitive Brain):** Operates the 4-Agent consensus mesh, analyzes symptoms, cites ICMR clinical treatment protocols, computes welfare benefits (JSY, PMMVY, PM-JAY), and enforces Six-Sigma safety vetoes.
- Swytchcode (The Execution Hand):** Enforces zero-trust tool execution, deduplicates emergency dispatches with SHA-256 sliding-window idempotency, and securely proxies Indic voice calls via Sarvam AI.

4. Live Production Verification & Telemetry (HTTP 200 OK)

Mesh Evaluation Metric	Live Response Telemetry across 4 Deployed Lyzr Agents
Endpoint Status	200 OK on GET /api/lyzr/status, GET /api/lyzr/agents, POST /api/lyzr/triage
Triage Urgency Result	CRITICAL (Pre-eclampsia danger sign triggered with BP 160/100 mmHg)
Safety Guardrail Veto	VETO TRIGGERED: TRUE (Severe hypertension flagged; immediate physical PHC referral mandated)
Official Citations	MoHFW Management of Hypertensive Disorders in Pregnancy & ICMR Standard Treatment Workflows
Welfare Entitlements	Janani Suraksha Yojana (JSY: INR 1,400) + PMMVY (INR 5,000 DBT) + Ayushman Bharat PM-JAY (INR 5,00,000)
Idempotent Tool Hand-off	Dispatched to Swytchcode runtime for zero-duplicate ambulance and ASHA worker alert

5. Mounted Backend Integration Endpoints

- GET /api/lyzr/status: Live status and configuration of all 4 deployed agents.
- GET /api/lyzr/agents: Complete multi-agent mesh topology listing roles and IDs.
- POST /api/lyzr/triage: Runs manager agent triage audited by the safety guardrail.
- POST /api/lyzr/schemes: Direct welfare entitlement calculation via Agent 3.
- GET /health: System-wide integration health check reporting verified provider statuses.